Week : 10

الأسبوع 10: من - إلى

Subject/:الموضوع Chemistry (IB DP) **Unit Title : عنوان الوحدة Reactivity 3.1 – Proton transfer reactions** **Grade: 12 (DP2)**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 **R3.1.1–3.1.3 Acids, bases and conjugate pairs**	**Learning target (DOK 2):** I can deduce Brønsted–Lowry acids, bases and conjugate acid–base pairs in a reaction, and formulate equations for species that act as both acids and bases.	**UDL strategies:** 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	IB Chemistry Interactive: PhET Acid-Base Solutions simulation – see how acids and bases behave in water. Class Teams (Reactivity 3.1 folder): lesson slides and conjugate-pair practice questions.
Lesson 2 **الحصة الثانية** **R3.1.4 The pH scale**	**Learning target (DOK 2):** I can perform calculations using pH = $-\log_{10}[H^+]$ and $[H^+] = 10^{(-pH)}$ and explain how pH describes the hydrogen ion concentration of a solution.	**UDL strategies:** 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Interactive: PhET pH Scale simulation – see how pH relates to $[H^+]$ in everyday solutions. Class Teams (Reactivity 3.1 folder): lesson slides and pH calculation practice. IB data booklet: the pH equations are given.
Lesson 3 **الحصة الثالثة** **R3.1.5 Ion product of water, Kw**	**Learning target (DOK 2):** I can use $K_w = [H^+][OH^-]$ to explain the inverse relationship between $[H^+]$ and $[OH^-]$ and to recognise solutions as acidic, neutral or basic.	**UDL strategies:** 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Interactive: PhET Acid-Base Solutions simulation – compare $[H^+]$ and $[OH^-]$ in different solutions. Class Teams (Reactivity 3.1 folder): lesson slides and Kw practice questions. IB data booklet: Kw and its value at 298 K are given.
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	Bring a calculator, Chemistry notebook, ruler, pencil, and coloured pens. Participate actively, complete all classwork, correct misconceptions, and upload required work to Toddle by the stated deadline		

Week 10 الأسبوع 10

Technology Integration دمج التكنولوجيا	Videos posted on Teams; PhET Acid-Base Solutions and pH Scale simulations; pH meters/probes and universal indicator; mini whiteboards; scientific calculators; digital IB data booklet.
Cross-curricular links الروابط بين المواد الدراسية	**Mathematics:** logarithms, powers of ten and rearranging equations. **Biology:** pH and enzyme activity. **Geography:** acid rain and its effects.
External Assessment Integration دمج التقييمات الخارجية	IB DP Chemistry Papers 1 and 2: deducing conjugate pairs, pH and Kw calculations, and data-booklet use, with the command terms deduce, calculate, determine and state.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Thinking:** applying logarithmic relationships. **Communication:** writing equations with state symbols. **Self-management:** rating themselves 1–5 against the success criteria.
Standards المعايير	• **R3.1.1** Brønsted–Lowry acid is a proton donor and a Brønsted–Lowry base is a proton acceptor. • **R3.1.2** A pair of species differing by a single proton is called a conjugate acid–base pair. • **R3.1.3** Some species can act as both Brønsted–Lowry acids and bases. • **R3.1.4** The pH scale can be used to describe the $[H^+]$ of a solution: $pH = -\log_{10}[H^+]$; $[H^+] = 10^{\wedge}(-pH)$. • **R3.1.5** The ion product constant of water, Kw, shows an inverse relationship between $[H^+]$ and $[OH^-]$. $Kw = [H^+][OH^-]$.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Guiding question:** What happens when protons are transferred? **Nature of science, Reactivity 3.4:** Why has the definition of acid evolved over time? **Nature of science, Tool 2:** When are digital sensors (e.g. pH probes) more suitable than analogue methods (e.g. pH paper/solution)?
Success Criteria معايير النجاح **(Measurable)**	● **Must (DOK 1–2):** identify Brønsted–Lowry acids, bases and conjugate pairs, and state the pH equation. ■ **Should (DOK 2–3):** calculate pH and $[H^+]$, and use Kw to classify solutions as acidic, neutral or basic. ▲ **Could (DOK 3–4):** formulate equations for species that act as both acids and bases, and justify why water ionizes more as temperature rises. **Evidence:** hinge questions, exit tickets and mini-whiteboard work.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء — Global Understanding التفاهم العالمي — Heritage التراث
	المجال Area: Innovation and Infrastructure
	التطبيق Implementation: Students link acids and bases to UAE contexts: acid rain from nitrogen and sulfur oxides, and pH monitoring of seawater and desalinated water.